

Cristiano Capone

cristiano0capone@gmail.com

The Editors, *Brain*

Dear Editors,

We are pleased to submit our manuscript, “Optimal stimulation sites are not the most affected: personalised fMRI models of Alzheimer’s disease,” by Cristiano Capone, Enza Cece, Luca Falorsi, Andrea Ciardiello, Guido Gigante, Evaristo Cisbani and Maurizio Mattia, for consideration in *Brain*.

Neuromodulation is being actively pursued in Alzheimer’s disease, and every such intervention requires a decision that is currently made on anatomical grounds: which site to stimulate. Targets have been selected where degeneration and network disruption are greatest, and clinical results have been equivocal. Whether “where the disease is worst” is the right selection criterion has never been tested causally, because doing so requires asking a counterfactual question of an individual patient’s brain.

We address this with personalised generative models. For each participant we fit a subject-specific, cross-subject-identifiable model whose free-running dynamics reproduce that individual’s own resting-state activity, and we then intervene on it *in silico*. Diagnostic labels come from ADNI’s own diagnosis query and are independently corroborated by cognition (MMSE 28.7 ± 1.4 in controls versus 22.7 ± 2.8 in patients, $p = 1.6 \times 10^{-13}$), and the discriminant is unrelated to age, education or sex. Restoring a virtual patient’s connectivity toward the control template reverts the disease signature, establishing that it is correctable in principle, but the required correction is distributed: no single-node edit reproduces it.

The clinically relevant result concerns where a focal drive should act. A single-site drive at the node whose connectivity is most altered fails to revert the classification at any amplitude, whereas a drive at the site selected by its *effect on the disease discriminant* reverts almost every patient. The two site sets are essentially disjoint, sharing one parcel out of 29 and 25, and none at all when the comparison is restricted to cortex. The reason is mechanistic: the most-affected nodes are the most weakly connected, and therefore poor actuators, while therapeutic leverage sits at well-connected cortical nodes that can redistribute a corrective perturbation. Targets are heterogeneous across patients, and a real-time closed-loop controller reaches comparable efficacy at substantially lower dose using only causally available information.

We are explicit about scope. This is an in-silico study; no patient was stimulated, and complete reversal requires amplitudes above the physiological range. Its contribution is a principle for target selection and a computable, per-patient prescription: if the effective target is systematically not the most affected region, then anatomically guided targeting may be mis-specified, and target choice should be model-informed and individualised. We believe this speaks directly to how neuromodulation trials in Alzheimer’s disease are designed, and therefore to the readership of *Brain*.

The manuscript is original, is not under consideration elsewhere, and all authors have approved its submission. The data derive from the Alzheimer’s Disease Neuroimaging Initiative and are used under its Data Use Agreement. We declare no competing interests, and have suggested potential referees who have no conflict of interest with the authors. We thank you for considering our work.

Sincerely,

Cristiano Capone, on behalf of all authors